Write a Python Program to Implement Expectation & Maximization Algorithm

CODE:

```
import numpy as np
from sklearn.mixture import GaussianMixture

X = np.array([
    [1.0], [1.5], [2.0], [2.5],
    [8.0], [8.5], [9.0], [9.5]
])
model = GaussianMixture(
    n_components=2,
    random_state=42
)
model.fit(X)
labels = model.predict(X)
print("Expectation-Maximization (EM) Algorithm")
print("----------------------------------------")

print("\nData Points:")
print(X.flatten())

print("\nCluster Labels:")
print(labels)

print("\nCluster Means:")
print(model.means_.flatten())
```

```
print("\nCluster Variances:")
print(model.covariances_.flatten())

print("\nMixing Probabilities:")
print(model.weights_)

print("\nLog Likelihood:")
print(model.lower_bound_)
```

OUTPUT:

```
Expectation-Maximization (EM) Algorithm
----------------------------------------

Data Points:
[1.  1.5 2.  2.5 8.  8.5 9.  9.5]

Cluster Labels:
[0 0 0 0 1 1 1 1]

Cluster Means:
[1.75 8.75]

Cluster Variances:
[0.312501 0.312501]

Mixing Probabilities:
[0.5 0.5]

Log Likelihood:
-1.5305103088643377
```